Inkwell Default Template Demo

Inkwell

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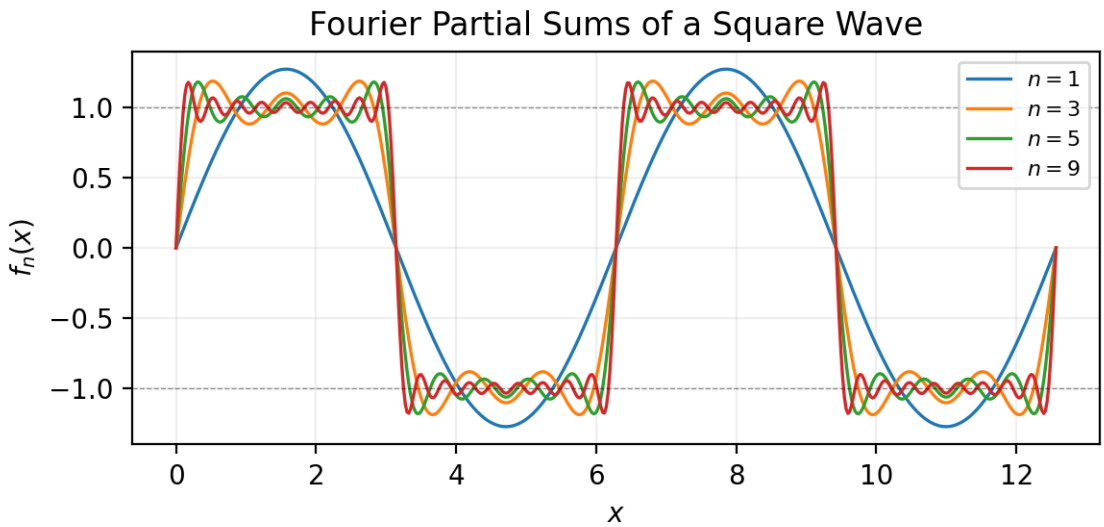

Figure 1: Fourier partial sums of a square wave for n = 1, 3, 5, 9.

1 Introduction

This document demonstrates the default Inkwell template. It compiles markdown to publication-quality PDF through Pandoc ([]) and XeLaTeX, with code blocks that execute in place. The result is a literate programming workflow where analysis and writing coexist in a single file ([]).

1.1 Runnable Code: Fourier Partial Sums

The Fourier series of a square wave ([]) converges pointwise but exhibits overshoot at discontinuities. The partial sum is:

$$f_n(x) = \sum_{k=1}^{n} \frac{4}{(2k-1)\pi} \sin\big((2k-1)x\big) \tag{1}$$

1.2 Inline Code with Output

Generate a scatter plot directly in the document using NumPy ([]) and Matplotlib ([]):

```
import os, numpy as np
import matplotlib; matplotlib.use("Agg")
import matplotlib.pyplot as plt

rng = np.random.default_rng(42)
```

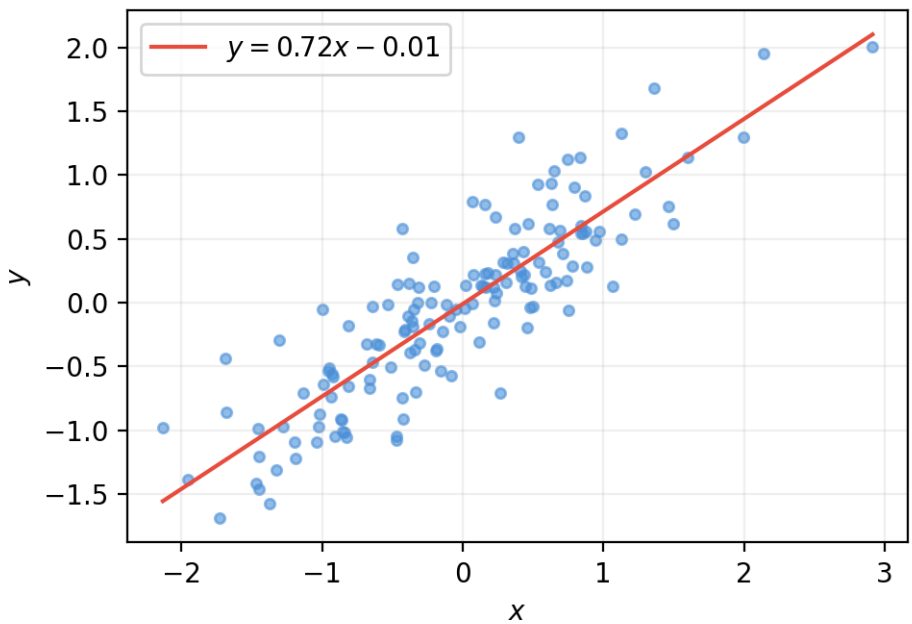

Figure 2: Simulated bivariate data with OLS regression line.

```
x = rng.normal(0, 1, 150)
y = 0.7 * x + rng.normal(0, 0.35, 150)

fig, ax = plt.subplots(figsize=(5, 3.5))
ax.scatter(x, y, s=14, alpha=0.6, color="#4A90D9")
m, b = np.polyfit(x, y, 1)
xs = np.sort(x)
ax.plot(xs, m * xs + b, color="#E74C3C", linewidth=1.5,
        label=f"$y = {m:.2f}x {'+' if b >= 0 else ''}{b:.2f}$")
ax.set_xlabel("$x$"); ax.set_ylabel("$y$")
ax.legend(); ax.grid(alpha=0.2)
fig.tight_layout()
fig.savefig(os.path.join(os.environ.get("INKWELL_OUTPUT_DIR", "."),
            "scatter.png"), dpi=200, bbox_inches="tight")
plt.close(fig)
print(f"n = {len(x)}, r = {np.corrcoef(x, y)[0,1]:.3f}")
```

1.3 Tables

Static markdown tables work as expected:

Table 1: Comparison of three methods on the benchmark dataset.

Method	Time (ms)	Accuracy (%)
Baseline	12.5	91.2
Proposed	10.1	93.8
Optimized	8.7	94.4

Tables can also be generated from code. A block that writes a `.csv` to the output directory is rendered as a formatted table:

Table 2: Descriptive statistics for the simulated bivariate data.

Variable	n	Mean	Std	Min	Max
x	150	-0.051	0.853	-2.132	2.914
y	150	-0.046	0.710	-1.690	2.003

1.4 Math

Euler's identity ([]) connects five fundamental constants:

$$e^{i\pi} + 1 = 0$$

A theorem environment rendered by the template:

Cauchy-Schwarz Inequality. For all vectors u, v in an inner product space,

$$|\langle u, v\rangle|^2 \leq \langle u, u\rangle \cdot \langle v, v\rangle$$

1.5 Environment

```
echo "Date: $(date '+%Y-%m-%d %H:%M')"
echo "Host: $(hostname -s), $(uname -s) $(uname -m)"
echo "Python: $(python3 --version 2>&1)"
echo "Pandoc: $(pandoc --version 2>/dev/null | head -1 || echo 'not installed')"
```

```
Date: 2026-02-22 21:50
Host: Elis-MacBook-Pro, Darwin arm64
Python: Python 3.9.6
Pandoc: pandoc 3.9
```

References

[] Euler, Leonhard. 1748. *Introductio in Analysin Infinitorum*.

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[] Harris, Charles R. et al. 2020. "Array Programming with NumPy." *Nature* 585: 357–62. https://doi.org/10.1038/s41586-020-2649-2.

[] Hunter, John D. 2007. "Matplotlib: A 2D Graphics Environment." *Computing in Science & Engineering* 9 (3): 90–95. https://doi.org/10.1109/MCSE.2007.55.

[] Knuth, Donald E. 1984. "Literate Programming." *The Computer Journal* 27 (2): 97–111. https://doi.org/10.1093/comjnl/27.2.97.

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